03/11/25

Definition (Measure zero sets)

Let E be a subset of $\mathbb{R}$ we say E is of measure zero, if for every $\epsilon > 0$ there exists a countable family of open intervals $\{I_j\}_{j=1}^{\infty}$ such that

$$E \subset \bigcup_{j=1}^{\infty} I_j \text{ and } \sum_{j=1}^{\infty} \Delta I_j \leq \epsilon$$

Examples:

(i) $E = \{a_1, \ldots, a_n\}$

$|E|$ = measure of E,

then $|E| = 0$

(ii) Any countable set is of measure zero

(iii) Any countable union of countable set is a measure zero.

Proof of (ii) $E = \bigcup_{j=1}^{\infty} \{x_j\}$

Given $\epsilon > 0$

Take $\left(x_j - \frac{\epsilon}{2^j}, x_j + \frac{\epsilon}{2^j}\right) = I_j$

$$E \subset \bigcup_{j=1}^{\infty} I_j$$

$$\Rightarrow \sum_{j=1}^{\infty} \Delta I_j \leq \sum_{j=1}^{\infty} \frac{1}{2^{j-1}}$$

$$< \epsilon.$$

Ex-(iii)

Example :

(1) Cantor set

↳ uncountable set.

↳ length of $[0,1] \setminus C$ is 1.

Theorem (Riemann-Lebesgue)

Let $f : [a,b] \to \mathbb{R}$ be bounded.

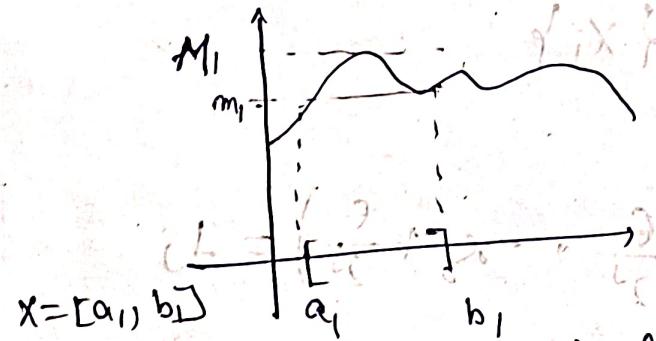
Then f is Riemann integrable

iff $\left[D(f) := \{ x \in [a,b] : x \text{ is a point of discontinuity of } f \} \right]$

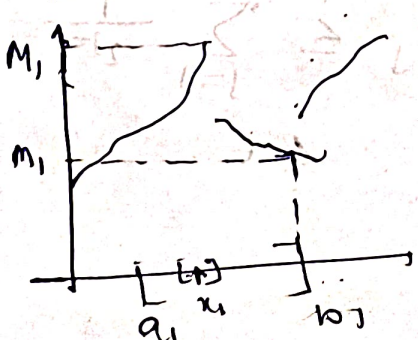
$D(f)$ is of measure zero.

→ ⊙ Denote

$$\operatorname{osc}_x(f) = \limsup_{t \to x} f(t) - \liminf_{t \to x} f(t)$$



$$\operatorname{osc}_x(f) = \sup_{[a_1,b_1]} f - \inf_{[a_1,b_1]} f = M_1 - m_1$$



$$osc_x f = \lim_{r \to 0} \left[\sup_{y \in [x-r, x+r]} f - \inf_{y \in [x-r, x+r]} f \right]$$

- If $x \notin D(f)$
  $osc_x(f) = 0$!! (because at x, f is continuous)
- If $x \in D(f)$
  $osc_x(f) \neq 0$!!

Result $x \notin D(f)$ iff $osc_x(f) = 0$

proof of the theorem :

Define set

$$D_K = \{ x \in [a,b] : osc_x f > \frac{1}{K} \}$$

Note that,

$$D(f) = \bigcup_{K=1}^{\infty} D_K$$

Note that $D(f)$ is of measure zero iff D_K are measure zero set $\forall K \in \mathbb{N}$

$\Rightarrow$ Assume f is Riemann Integrable.

Claim : D_K are of measure zero $\forall K \in \mathbb{N}$.

Let $\epsilon > 0$, given any $\epsilon \in \mathbb{N}$ there is a partition P such that

$$U(P,f) - L(P,f) < \epsilon/K$$

$$\Rightarrow \sum (M_i - m_i) \Delta I_i < \epsilon/K$$

$D_K \subset [a,b]$

Choose $\{I_{K_1}, \ldots, I_{K_{n_0}}\}$ such that

$$D_K \cap I_{K_j} \neq \phi \quad \forall j \in \{K_1, \ldots, K_{n_0}\}$$

Let $x \in D_K$.

By the definition of D_K we have,

$(M_j - m_j) \geq \frac{1}{K}$

(Note $I_j \cap D_K \neq \emptyset$

Hence $x \in I_j \cap D_K$

$M_j - m_j \geq \frac{1}{K}$ (Because $x \in I_j$)

Now, $\epsilon > \sum_{j=1}^{n} (M_j - m_j) \Delta x_j$

$\geq \sum_{j=1}^{n_0} (M_j - m_j) \Delta x_j$

$\geq \frac{1}{K} \sum_{j=1}^{n_0} \Delta x_j$

$\Rightarrow \sum_{j=1}^{n_0} \Delta x_j < \epsilon$

and $D_K \subseteq \bigcup_{j=1}^{n_0} I_j$

Consider open intervals

$\tilde{I}_j \supset I_j$

such that

$\sum_{j=1}^{n_0} |\tilde{I}_j| < 2\epsilon$.

As $\epsilon > 0$ is arbitrary

$\Rightarrow D_K$ is a measure zero set.

($\Leftarrow$) Assume $D(f)$ is measure zero

$\Rightarrow D_K$ are measure zero $\forall K \in \mathbb{N}$.

Let $D_{1/\epsilon} = \{x \in [a,b];\ osc_x f > \epsilon\}$

↓

measure zero

Then $\exists$ countable union of I_j such that

$$D_{1/\epsilon} \subset \bigcup_j I_j \text{ and } \sum \Delta I_j < \epsilon$$

Note (Exercise):

$D_{1/\epsilon}$ is a compact set.

then there is a finite subcover

$$D_{1/\epsilon} \subseteq \bigcup_{j=1}^{N} I_j \text{ and } \sum_{j=1}^{N} \Delta I_j < \epsilon.$$

Consider $K = [a,b] \setminus \bigcup_{j=1}^{N} I_j$

Note K is a compact set

$$K \subset [a,b] \setminus D_{1/\epsilon}$$

**

Aim: $\exists$ $P \to$ partition

$$U(P,f) - L(P,f) < \epsilon$$

Construct a partition which contains D_ϵ (→ bad set) and $[a,b] \setminus D_\epsilon$ (→ good set)

Separately

$$U(P,f) - L(P,f) \leq \sum_{\text{bad}} (M_i - m_i)\Delta I_i + \sum_{\text{good}} (M_j - m_j)\Delta I_j$$

$$\leq 2M \underbrace{\sum_{\text{bad}} \Delta I_i}_{\leq \epsilon} + \epsilon \sum_{\text{good}} \Delta I_j$$

for any $y \in [a,b] \setminus D_{1/\epsilon}$

$\exists\ I_y$ open interval such that

$y \in I_y \subseteq [a,b] \setminus D_{1/\epsilon}$

~~$K \setminus D_{1/\epsilon} = \bigcup I_y$~~

Then $K \subseteq \bigcup_y I_y$

K compact

$\Rightarrow \exists\ J_i$ s.t

$$K \subseteq \bigcup_{i=1}^{n_0} J_i$$

Also note that $J_i \subseteq [a,b] \setminus D_{1/\epsilon}$

we get $\sup_{x,y \in J_i} |f(x) - f(y)| \leq \epsilon$. (check!)

Now, create a partition containg end points of $\{I_1, \ldots, I_n\}$ and $\{J_1, \ldots, J_{n_0}\}$